

Stock-Market Dynamics, Rev. 5: Honest Negative on the Return Mechanism, Data Integrity Fix, and Three Risk Metrics

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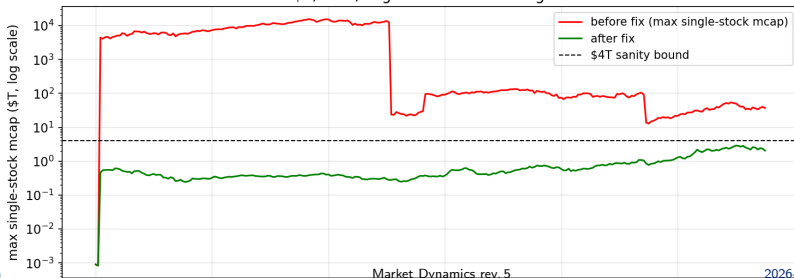
Rev-5 in one slide

- ▶ Rev-4's diagnosis: the model predicts *where capital sits*, not *which bins will pay* (mass-flow $\rightarrow$ return corr. 0.009)
- ▶ Rev-5 asked: does concentration predict returns? **Pre-registered test $\rightarrow$ honest negative.** No return mechanism built.
- ▶ Along the way: found and fixed **consistent** 10^3 – $10^6 \times$ **unit errors** in share counts (SRE backfill = \$5,000T) that corrupted every cap-weighted object in rev-4
- ▶ Three **risk metrics** as diagnostics: ensemble forecast intervals, fragility index, portfolio risk
- ▶ Clean-data re-calibration: distribution hindcasts improve markedly
- ▶ **No new paper track. No real-capital deployment.**

Data integrity fix

- ▶ Jump-based cleaner missed *consistent* wrong units: SRE backfill \$5,000T $\rightarrow$ top-lane share flipped 0.999/0.002 monthly; QCOM filed era \$136T; EOG, SYK, HAL, Citi ...
- ▶ Fix: segment split-adjusted shares at $> 20\times$ jumps; check each segment vs. **absolute anchors** — cap in [\$200M, \$4T], daily turnover in [0.02%, 100%]
- ▶ Rescale by $10^{\pm 3}$, $10^{\pm 6}$ only where *both* turn sane; drop (never invent) the rest: 28 segments fixed, 17 dropped
- ▶ Verification: binnable cap 2007 \$10.2T ($\approx$ \$13T true), 2021 \$41.3T ($\approx$ \$40T)

Data integrity fix: consistent unit errors in filed share counts
SRE backfill read \$5,000T; segment-level rescaling to absolute anchors

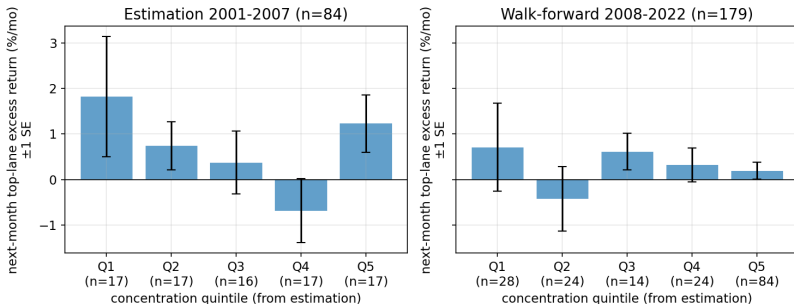


Concentration → return: honest negative

Pre-registered, estimated 2001–2007, judged walk-forward 2008–2022:

- ▶ Form 1 (threshold-linear): $b = -0.18$, $t = -0.41$, $R^2 = 0.002$; WF corr -0.045 — **fail**
- ▶ Form 2 (quintiles): Q5–Q1 $-0.59\%/mo$ ($t = -0.40$) in-sample, $-0.52\%/mo$ ($t = -0.52$) walk-forward — **statistically zero**, non-monotone
- ▶ $\text{corr}(C, r^e)$: -0.083 / -0.001 . Nothing tradeable.

Concentration → return test: no monotone crowding penalty (honest negative)
Q5–Q1 spread: in-sample -0.59% ($t=-0.40$), walk-forward -0.52% ($t=-0.52$)

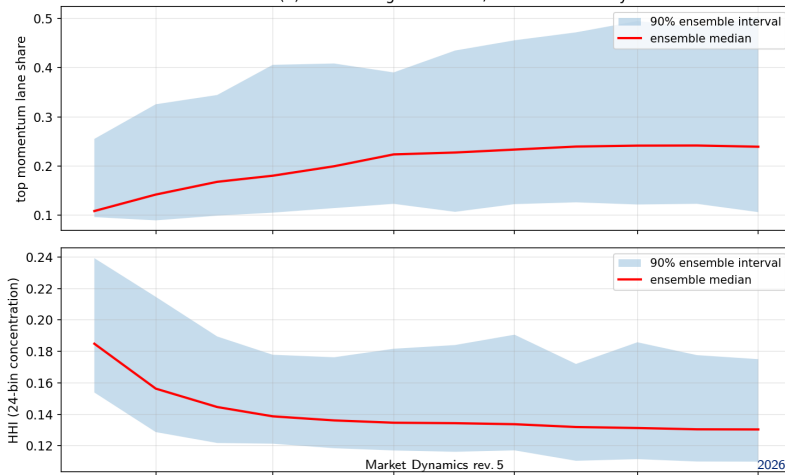


Risk metric 1: ensemble forecast with intervals

200 scenarios from 2022-12: Dirichlet T (Kish N_{eff}) + AR(1) driver + VIX-regime Markov:

- ▶ 12-mo top-lane share: median **0.239**, 90% interval **[0.106, 0.503]**
- ▶ HHI: [0.110, 0.175]. The width *is* the finding.

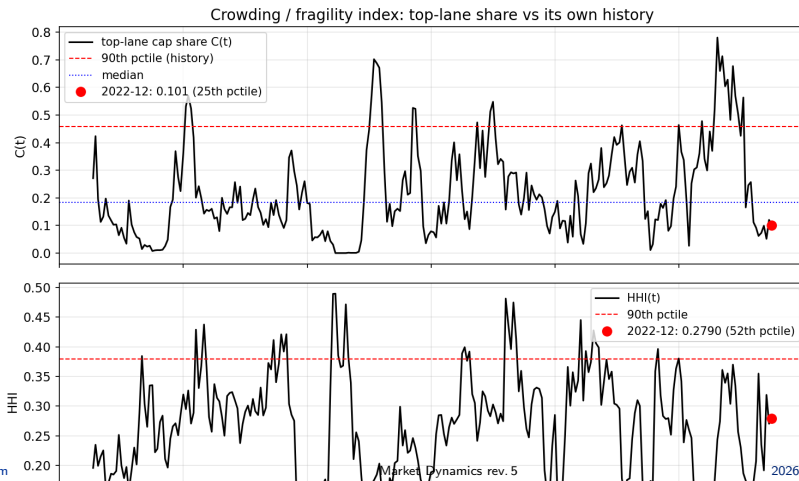
Rev-5 ensemble distribution forecast from 2022-12 (200 scenarios)
Dirichlet-T + AR(1)-M + VIX-regime Markov; intervals on every forecast



Risk metric 2: fragility index

Top-lane share and HHI as percentiles of own history:

- ▶ **2022-12: $C = 0.101$ (25th pctl — not crowded);** HHI 52nd pctl
- ▶ Most crowded: 2004-02 (0.570), 2010-01 (0.571), 2001-02 (0.423)
- ▶ A state descriptor, **not** a signal — crowding does not predict returns (§3)



Risk metric 3: portfolio risk

- ▶ Predicted vol from 24-bin covariance (2001–2022): tilt **14.4%** vs. market 15.9% vs. SPY 15.4% — the tilt's failure was return-side, not risk-side
- ▶ $P(\text{top lane unwinds next month} \mid C \text{ quintile})$: 41.8% / 44.0% / 35.8% / 53.8% / **34.0%** — no monotone pattern; most-crowded reverses *least*

Clean-data hindcasts (realized- M forcing): 2008–09 RMSE **0.0899** (was 0.1685); 2020–21 **0.0714** (was 0.1110), peak amplitude exact. The distribution mechanics survive the fix and work better.

No return mechanism. No new paper track. No real-capital deployment.

- ▶ Concentration does not predict returns (honest negative, $n = 263$ months)
- ▶ Risk metrics are **diagnostics**, not signals: ensemble intervals, fragility percentiles, predicted vol / reversal odds
- ▶ Distribution model re-calibrated on clean data; code and data preserved for any future attempt
- ▶ Next: return-side economics (earnings, flows) — or admit the framework prices *states*, not *returns*

References

- ▶ fja05680/sp500 — *S&P 500 Historical Components & Changes* (public GitHub reconstruction). Membership.
- ▶ Yahoo Finance — adjusted closes, splits, SPY total return. Prices, benchmark.
- ▶ SEC EDGAR — companyconcept XBRL shares outstanding. Point-in-time shares; consistent-unit-error repair documented.

Thank you.